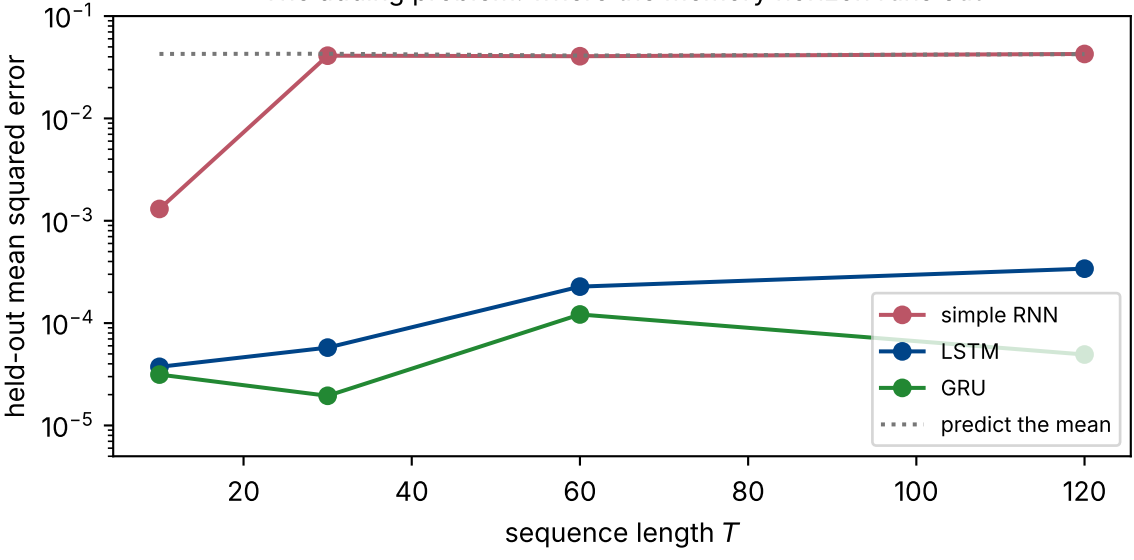
The adding problem: where the memory horizon runs out
held-out mean squared error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
20
40
60
80
100
120
sequence length T
simple RNN
LSTM
GRU
predict the mean